

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

C05: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications. (BTL 5)*

Assessment Tool Weightage: 50%

QUESTIONS: 5

Annexure A

Duration : 1 Hour
Total Marks:50

Design Task

Code of Conduct:

I M Sandhya(192312421) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

M Sandhya

Signature of the student:

Design Task

1. Hybrid AI Recommendation Engine Design

Design a hybrid AI-based recommendation system that integrates content-based and collaborative filtering techniques using PySpark RDDs. Explain how distributed data processing can be used to handle large-scale user-item interactions, and describe the role of intelligent agents in adapting recommendations based on user behavior and feedback in real time.

2. Smart Disaster Detection System

Develop a distributed AI system using PySpark and RDD operations to detect natural disasters (such as floods or earthquakes) from multi-source data streams (satellite images, sensors, and social media). Describe the system architecture, data fusion techniques, and how intelligent agents coordinate to provide early warnings and response strategies.

3. AI-Based Fraud Detection Framework

Design a scalable fraud detection system using PySpark RDDs to process large volumes of financial transactions. Explain how anomaly detection algorithms can be implemented in a distributed manner and how intelligent agents collaborate to identify suspicious patterns, trigger alerts, and continuously improve detection accuracy.

4. Autonomous Supply Chain Optimization System

Construct a distributed AI system using PySpark for optimizing supply chain operations (inventory, logistics, demand forecasting). Describe how RDD transformations support large-scale data processing and how multiple intelligent agents interact to make decentralized decisions for cost minimization and efficiency improvement.

5. Personalized Learning Analytics Platform

Design an AI-driven educational platform that uses PySpark and RDDs to analyze student learning behavior and performance data. Explain how the system supports adaptive learning through intelligent agents, real-time feedback, and scalable analytics to personalize content delivery for thousands of learners.

RUBRICS FOR CALCULATION

Evaluation Criteria	Problem Understanding & Requirement Analysis	System Architecture Design	Use of PySpark & RDD Operations	Intelligent Agent Integration	Scalability & Distributed Computing Design	Data Flow & Processing Pipeline	Innovation & Creativity	Clarity, Presentation & Documentation
Weightage	10%	20%	15%	15%	10%	10%	10%	10%

Criteria	Excellent (5)	Good (4)	Average (3)	Poor (2)
Problem Understanding & Requirement Analysis	Clearly defines problem, objectives, and constraints	Mostly clear with minor gaps	Basic understanding	Unclear or incorrect
System Architecture Design	Complete, scalable, well-structured architecture with diagrams	Good design with minor issues	Basic design, lacks detail	Poor/incomplete
Use of PySpark & RDD Operations	Efficient and correct use of RDD transformations/actions	Mostly correct usage	Limited or partial use	Incorrect/missing
Intelligent Agent Integration	Strong integration with clear agent roles and logic	Good integration	Basic integration	Poor/no integration
Scalability & Distributed Computing Design	Clearly addresses scalability, fault tolerance, parallelism	Covers most aspects	Limited consideration	Not addressed

Data Flow & Processing Pipeline	Clear, logical, and well-defined data flow	Mostly clear	Some gaps	Poorly defined
Innovation & Creativity	Highly innovative, realistic, and impactful solution	Some innovation	Basic idea	No originality
Clarity, Presentation & Documentation	Very clear, neat diagrams, well-organized report	Mostly clear	Somewhat organized	Poor presentation

ANSWERS

1. Hybrid AI Recommendation Engine Design

1. Objective

Design a scalable hybrid recommendation system integrating content-based filtering and collaborative filtering using PySpark RDDs.

2. System Architecture

Data Sources → Preprocessing → RDD Creation → Model Computation → Recommendation Engine → Feedback Loop

3. RDD-Based Processing

- Create RDDs:
 - `UserRDD (user, item, rating)`
 - `ItemRDD (item, features)`
- Apply transformations:
 - `map()` → structure data
 - `groupByKey()` → user profiles
 - `join()` → combine user & item data
 - `reduceByKey()` → similarity computation

4. Hybrid Recommendation Model

- Content-based: cosine similarity between user profile & item features
- Collaborative: user-user or item-item similarity
- Final score:

Score = $\alpha(\text{Content}) + \beta(\text{Collaborative})$

5. Distributed Processing

- Parallel similarity calculations
- Efficient handling of large-scale datasets
- Fault tolerance using RDD lineage

6. Intelligent Agents

- User Agent: monitors interactions
- Recommendation Agent: generates suggestions
- Learning Agent: updates model dynamically
- Feedback Agent: adjusts weights based on user feedback

7. Outcome

Accurate, scalable, and adaptive personalized recommendations.

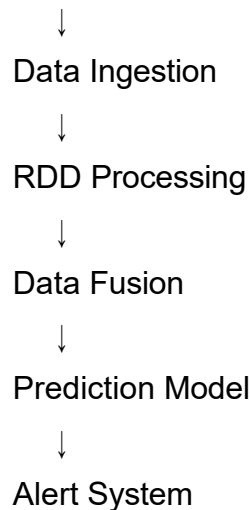
2. Smart Disaster Detection System

1. Objective

Develop a distributed AI system for detecting disasters using multi-source data streams.

2. System Architecture

Satellite Data + Sensors + Social Media



3. RDD Operations

- `mapPartitions()` → image processing
- `filter()` → remove noise
- `union()` → combine streams
- `reduceByKey()` → region-based aggregation

4. Data Fusion Techniques

- Feature-level fusion (sensor + image)
- Decision-level fusion (combine model outputs)
- Temporal fusion (time-series trends)

5. Intelligent Agents

- Sensor Agent: monitors environmental data
- Vision Agent: analyzes satellite images
- Social Agent: detects disaster-related posts
- Coordinator Agent: integrates results
- Alert Agent: issues warnings

6. Outcome

Real-time disaster detection and early warning system.

3. AI-Based Fraud Detection Framework

1. Objective

Design a scalable fraud detection system using PySpark RDDs for large transaction datasets.

2. System Architecture

Transaction Data → Preprocessing → Feature Extraction → RDD Processing → Anomaly Detection → Alert System

3. RDD Processing

- `map()` → transaction feature extraction
- `filter()` → remove invalid entries
- `groupByKey()` → user transaction history
- `reduceByKey()` → aggregate behavior patterns

4. Anomaly Detection Algorithms

- Statistical methods (Z-score)
- Machine learning models (Isolation Forest, clustering)
- Rule-based detection

5. Distributed Implementation

- Parallel processing of transactions
- Real-time fraud scoring
- Scalability across distributed nodes

6. Intelligent Agents

- Transaction Agent: monitors incoming transactions
- Detection Agent: identifies anomalies
- Alert Agent: flags suspicious activities
- Learning Agent: updates fraud patterns continuously

7. Outcome

Fast, scalable, and adaptive fraud detection system.

4. Autonomous Supply Chain Optimization System

1. Objective

Develop a distributed AI system for optimizing inventory, logistics, and demand forecasting.

2. System Architecture

Data Sources → RDD Processing → Forecasting Models → Optimization Engine → Decision Execution

3. RDD Transformations

- `map()` → preprocess supply chain data
- `groupByKey()` → warehouse-level data
- `reduceByKey()` → demand aggregation
- `join()` → integrate inventory & demand

4. Optimization Techniques

- Demand forecasting using ML models
- Inventory optimization algorithms
- Route optimization for logistics

5. Intelligent Agents

- Inventory Agent: manages stock levels
- Logistics Agent: optimizes transportation
- Demand Agent: predicts future demand
- Coordinator Agent: ensures system-wide optimization

6. Distributed Processing Role

- Real-time decision making across regions
- Efficient handling of large datasets
- Parallel optimization computations

7. Outcome

Reduced costs, improved efficiency, and optimized supply chain operations.

5. Personalized Learning Analytics Platform

1. Objective

Design an AI-based platform to analyze student behavior and provide adaptive learning experiences.

2. System Architecture

Student Data → Preprocessing → RDD Analytics → Learning Model → Personalization Engine → Feedback System

3. RDD-Based Processing

- `map()` → extract learning features
- `groupByKey()` → student profiles
- `reduceByKey()` → performance aggregation
- `join()` → combine content and student data

4. Adaptive Learning Mechanism

- Analyze performance trends
- Recommend personalized content
- Adjust difficulty levels dynamically

5. Intelligent Agents

- Student Agent: tracks learning activity
- Content Agent: manages learning materials
- Recommendation Agent: suggests content
- Feedback Agent: provides real-time guidance

6. Distributed Processing

- Supports thousands of learners simultaneously
- Real-time analytics on large datasets
- Scalable content delivery

7. Outcome

Personalized, scalable, and intelligent learning platform.